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Operations Research

Prof. Dr. Martin Bichler,
Halil Bayrak, Janik Bürgermeister, Teodora Dobos,
Jan Höhener, Abheek Ghosh

Exercise Sheet 11

Demo Exercises

Exercise D11. 1. *Mirror Descent and Bregman Divergence*

Mirror descent generalizes gradient descent by replacing the Euclidean distance with a Bregman divergence $D_h(x\|y)$ generated by a strictly convex function $h(x)$. The update rule in the primal space is given by solving:

$$x^{(k+1)} = \arg\min_x \left\{ \langle \eta \nabla f(x^{(k)}), x \rangle + D_h(x\|x^{(k)}) \right\}$$

which inherently leads to the dual update: $\nabla h(x^{(k+1)}) = \nabla h(x^{(k)}) - \eta \nabla f(x^{(k)})$.

(a) Let $h(x) = \frac{1}{2}\|x\|_2^2$. Compute the Bregman divergence $D_h(x\|y)$ and show that the Mirror Descent update reduces to standard Gradient Descent.

(b) Let $h(x) = \sum_{i=1}^n (x_i \log x_i - x_i)$. Compute the gradient $\nabla h(x)$ and its inverse $\nabla h^{-1}(y)$.

(c) Using the function h from (b), derive the coordinate-wise update rule for $x_i^{(k+1)}$. How does this relate to the Hedge (Exponential Weights) algorithm?

Exercise D11. 2. *Projected Gradient Descent*

Consider minimizing a convex, differentiable function $f(x)$ over $X = \{x \in \mathbb{R}^2 \mid \|x\|_2 \leq 1\}$.

(a) Formulate the update rule for Projected Gradient Descent using $\Pi_X(\cdot)$.

(b) Give an explicit formula for the Euclidean projection operator $\Pi_X(y)$ for any point y.

(c) Let $f(x) = (x_1 - 2)^2 + (x_2 - 2)^2$. Perform one step starting at $x^{(0)} = (0, 0)^T$ with $\eta = 0.5$.

Exercise D11. 3. *From Follow-The-Leader to Regularization (FTRL)*

Consider an online linear optimization problem over the decision space $X = [-1, 1]$. In each round t, the adversary reveals a loss function $L_t(x) = g_t x$. The adversary plays the sequence of gradients: $g_1 = 0.5, g_2 = -1, g_3 = 1, g_4 = -1$.

(a) **The Failure of FTL:** The Follow-The-Leader algorithm chooses $x_t = \arg\min_{x \in X} \sum_{i=1}^{t-1} g_i x$ (with $x_1 = 0$). Compute the sequence of decisions x_t, the cumulative loss of FTL, and the overall regret compared to the best fixed decision in hindsight x^*.

(b) **The "Cheating" Baseline:** The hypothetical Be-The-Leader (BTL) strategy cheats by looking one step into the future: $x_t^{BTL} = \arg\min_{x \in X} \sum_{i=1}^{t} g_i x$. Compute the decisions and the regret of the BTL strategy. What does the difference between FTL and BTL reveal about the nature of adversarial regret?

(c) **The Cure (FTRL):** To prevent the wild oscillations of FTL, we introduce Follow-The-Regularized-Leader with a strongly convex penalty $h(x) = \frac{1}{2}x^2$ and a regularization weight $\eta^{-1} > 0$. The update rule becomes:

$$x_{t+1} = \arg\min_{x \in X} \left(\sum_{s=1}^{t} g_s x + \frac{1}{2\eta} x^2 \right)$$

Derive the closed-form expression for x_{t+1} as a function of the historical gradients and η. Briefly discuss how this relates to Online Gradient Descent (OGD) and prevents the failure observed in part (a).